

Genome-wide identification and protein interactions of the TIFY family in the bioenergy plant *Jatropha curcas* L.

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ABSTRACT

Jatropha curcas L. is an important energy crop whose growth is affected by abiotic stress. TIFY proteins regulate plant growth and respond to adverse conditions in higher plants. However, little information is available regarding the functions of the TIFY family genes in *J. curcas*. In this research, 14 members of the JcTIFY family were identified in *J. curcas* at the genome-wide level. These members belong to four subfamilies: jasmonate-ZIM-domain (JAZ), ZIM/ZIM-like, PEAPOD, and TIFY, and they are distributed across seven linkage groups. The TIFY proteins, consisting of 135–458 amino acids, are all hydrophilic and are predicted to be located in the nucleus. Although members of the same branch have different exon-intron structures, they share similar conserved domains and motif compositions. Protein interaction predictions indicate that TIFY proteins can form dimers and are involved in the jasmonic acid and ethylene signaling pathways. The role of TIFY in the growth of *J. curcas* was demonstrated by the analysis of cis-acting elements and expression of TIFY under various conditions and tissues. Furthermore, RT-qPCR analysis revealed that chilling stress and methyl jasmonate significantly induced the expression of JcJAZs, particularly JcTIFY5B. Yeast two-hybrid analysis revealed clear interactions between JcJAZs. Luciferase complementation and yeast two-hybrid assays demonstrated that JcTIFY10 formed a homodimer both *in vivo* and *in vitro*. JcJAZ proteins can form heterodimers with other proteins of the same family, and some proteins can form homodimers via the TIFY domain. The subcellular localization of JcTIFY5A and JcTIFY10 was in the nucleus, while JcTIFY9 was found in both the cytoplasm and nucleus. These results provide significant insights into the TIFY gene-mediated abiotic stress response and plant hormone signal transduction events in *J. curcas*.

1. Introduction

TIFY is a family of terrestrial plant genes involved in hormonal responses, environmental stimuli, and plant growth responses (Vanholme et al., 2007). TIFY was initially identified as a zinc-finger protein found in the inflorescence meristem of *Arabidopsis thaliana*, called ZIM (Vanholme et al., 2007). With advances in molecular biology, these proteins have been recognized as containing a highly conserved TIFY (Threonine, Isoleucine, Phenylalanine, Tyrosine) domain, leading to their naming as TIFY. TIFY proteins possess a highly conserved amino

acid sequence and are divided into four subfamilies based on other conserved sequence domains: TIFY, jasmonate-ZIM-domain (JAZ), ZIM/ZIM-like (ZML), and PEAPOD (PPD) (Chini et al., 2009; Galan et al., 2023). The TIFY subfamily is characterized solely by the presence of the TIFY domain, which is essential for protein-protein interactions (Galan et al., 2023). In addition to the conserved TIFY domain, ZML proteins include a GATA zinc finger domain and a CCT domain (Chini et al., 2009). JAZ proteins feature a conserved TIFY domain, a Jas domain with a conserved SLX2FX2KRX2RX5PY sequence, and a weakly conserved NT domain at the N-terminus. The Jas domain is annotated as

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a divergent CCT motif in NCBI. The NT, Jas, and TIFY domains mediate interactions between JAZ and various proteins (Hou et al., 2010; Chini et al., 2009). Lastly, the PPD proteins possess a conserved PPD domain located at the N-terminus and a Jas domain at the C-terminus that lacks a PY motif (Baekelandt et al., 2018).

Numerous investigations have demonstrated that TIFY proteins function in plant growth and are crucial for plants to respond to environmental and biological stresses. In *A. thaliana*, PPD proteins are involved in regulating leaf size and morphology (White, 2006). PPD is the only known transcriptional regulator that negatively regulates meristem division and is responsible for controlling the size and shape of organs in *A. thaliana* (Gonzalez et al., 2015). The PPD protein and the novel interactor of JAZ (NINJA) protein interact with the kinase-inducible domain-interacting proteins 8/9 (KIX8/9) and the corepressor TOPLESS (TPL), thereby negatively regulating the transcription of downstream genes such as *cyclin D3*, which control plant organ size (Gonzalez et al., 2015; Baekelandt et al., 2018; Swinnen et al., 2022). STERILE APETALA (SAP), which is a part of the SKP1/Cullin/F-box E3 ubiquitin ligase complex (SCF^{SAP} complex), can interact with PPD1/2 and KIX8/9 to degrade the KIX-PPD repressor complex, thereby promoting the proliferation of meristematic cells to control organ size (Wang et al., 2016). PPD proteins are also involved in plant responses to light intensity, with *AtPPD1* and *AtPPD2* transcripts being upregulated under low light and downregulated under high light. Their transcriptional activation is regulated by phytochrome interacting factor 4 (White, 2022). JAZ proteins function as repressive elements in the jasmonic acid (JA) signaling pathway (Song et al., 2022). JAZ proteins recruit TPL as a corepressor through the NINJA protein as an adaptor and bind to transcription factors to inhibit transcription activation (Pauwels et al., 2010). Excess AtJAZ levels render *A. thaliana* sensitive to low temperature stress (Hu et al., 2013). Compared with wild-type (WT) soybeans, transgenic plants overexpressing *GsJAZ2* exhibit enhanced alkali resistance and accelerated root growth under exogenous methyl jasmonate (MeJA) treatment (Zhao et al., 2020). Multiple *OsTIFYs* were overexpressed in rice, resulting in transgenic plants exhibiting an enhanced growth phenotype. The yield of these transgenic plants exceeded that of the WT under normal temperature or high temperature stress, and they were not sensitive to the inhibitory effects of JA (Hakata et al., 2017). Under salt stress, transgenic rice overexpressing *OsJAZ8* exhibited decreased mortality and sensitivity to MeJA, but increased sensitivity to auxin (Peethambaran et al., 2018). In *A. thaliana*, loss-of-function mutations in *JAZs* (1–7, 9, 10, 13) have been identified, and these mutants exhibited significant resistance to biotic stress; however, their vegetative growth was slow, and reproductive success was limited (Guo et al., 2018). Studies have demonstrated that the TIFY8 protein, a part of the TIFY subfamily of *A. thaliana*, is a transcriptional suppressor (Pérez et al., 2014) that interacts with the transcription factor REVOLUTA to inhibit the activation of senescence-related genes such as *WRKY53*. Overexpression of *AtTIFY8* in *A. thaliana* slowed the course of senescence, whereas loss-of-function mutations in *AtTIFY8* accelerated the course of senescence (Galan et al., 2023).

Jatropha curcas L. is a multipurpose tree that is part of the *Jatropha* genus within the Euphorbiaceae family. It is regarded as an excellent bioenergy plant due to its high oil content, and various components of *J. curcas* contain a variety of medicinal substances (Contran et al., 2013). It possesses important research value for pharmaceuticals (Igbinsola et al., 2011), industrial production (Ji et al., 2023; Premjet et al., 2021), improving the ecological environment, and purifying water resources (Garcia-Martin et al., 2020). However, the biomass and distribution of *J. curcas* are significantly constrained by various abiotic stresses, especially low temperature. The TIFY genes may play a crucial role in how *J. curcas* copes with these stresses; however, relevant studies on this gene family are scarce. In this study, 14 members of the TIFY family in *J. curcas* were identified through bioinformatics analysis. A phylogenetic tree was constructed, and their sequence characteristics, chromosome

localization, gene structure, cis-acting elements, and expression patterns were analyzed. The expression of the JAZ subfamily under low-temperature stress and exogenous MeJA treatment was studied. The subcellular localization of some JAZ proteins was verified in *Nicotiana benthamiana*. Finally, we demonstrated the interaction patterns of JcJAZ proteins using a yeast two-hybrid assay and luciferase complementation assay (LCA). This study provides a foundation for understanding the function of the TIFY gene and contributes to the investigation of the JA/MeJA signaling pathway in *J. curcas*.

2. Materials and methods

2.1. Plant material and treatments

Experimental seeds were purchased from Yuan Mou (Yunnan Province, China) and stored at room temperature. Moderately sized seeds were selected and cleaned with tap water. Their surfaces were disinfected with 1 % (m/v) copper sulfate for 20 min, followed by rinsing with ddH₂O and soaking in ddH₂O for 1 day. The seeds were spread flat in covered trays lined with filter paper and allowed to germinate in the dark at 26°C for 7 days. Each day, 50 mL of water was added to maintain the surface water layer (Fig. 1A). After the seeds took root, the seedlings were selected and transferred into white trays containing silica sand moistened with Hoagland's nutrient solution. The trays were placed in an artificial climate chamber with day/night temperatures set at 26°C/20°C, a photoperiod of 16 h light and 8 h dark, and a relative humidity of 75 % until the seedlings reached their third true leaf (Fig. 1B). The seedlings of uniform size were exposed to 5°C and 5°C + 60 μmol·L⁻¹ MeJA, with 75 mL of Hoagland's solution added per tray each day. The second and third true leaves of the seedlings were harvested at 0, 1, 2, 3, 4, and 5 days for subsequent experiments.

2.2. Identification of TIFY family genes in *J. curcas*

The *J. curcas* genome and protein data were downloaded from the NCBI website (<https://www.ncbi.nlm.nih.gov/>) (RJCI_Hi-C, BioProject: PRJNA601606), and two methods were used to identify TIFY genes within the whole genome of *J. curcas*. First, all schema data files from the Pfam database (<http://pfam.xfam.org/>) were downloaded, and the Hidden Markov Model (PF06200) of the TIFY genes was obtained. TBtools (V2.001) (Chen et al., 2020) was then used to search for candidate TIFY proteins in the *J. curcas* protein database. Second, all 18 TIFY proteins of *A. thaliana* (downloaded from <https://www.arabidopsis.org/>) were used as reference sequences to conduct a BLAST search in the *J. curcas* protein database (E-value < 10⁻⁵). Finally, candidate genes were obtained by filtering variable splice variants and were subsequently verified for conserved domains (CDs); thus, the TIFY family members (JcTIFYs) were identified. MEGA11 was utilized to conduct multiple sequence comparisons of *A. thaliana*, *Oryza sativa* (Ye et al., 2009), *Manihot esculenta* (Zheng et al., 2022), *Betula platyphylla* (Lv et al., 2023), and *J. curcas* TIFY proteins. A phylogenetic tree was generated using the neighbor-joining approach with the following parameter settings: P-distance model, complete deletion, and bootstrap analysis with 1000 replicates. Family members were named based on their evolutionary relationships with *A. thaliana* homologs.

2.3. Chromosome localization and prediction of physicochemical properties of the TIFY gene family in *J. curcas*

The gff3 annotation file for the *J. curcas* genome was downloaded from NCBI, and the chromosome location information of the genes was obtained by combining it with the high-density linkage map (Wu et al., 2015). The chromosome distribution of the genes was mapped using MapGene2Chrom (V2.0, <http://mg2c.iask.in/mg2c%5Fv2.0/>). Collinear analysis, based on local BLAST (parameter settings: E-value < 10⁻⁵, num_alignments 5), and gene location information of the protein

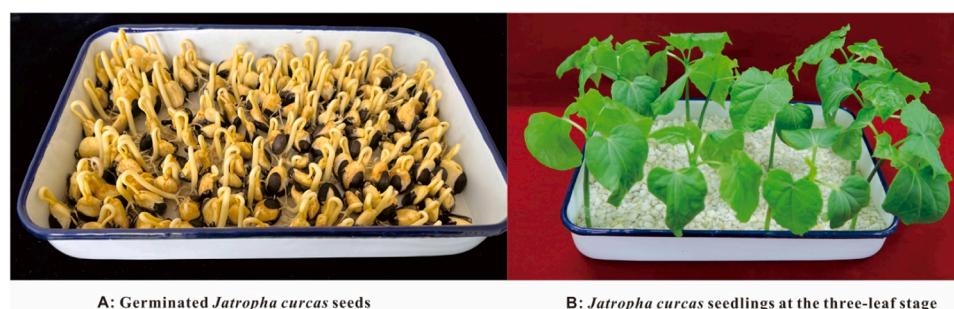


Fig. 1. Materials used in the experiment.

sequence were performed using the MCSan X program built into TBtools (V2.001). OrthoFinder was used to identify homologous gene pairs. The values for the non-synonymous substitution rate (Ka) and synonymous substitution rate (Ks) were calculated using MEGA7 software based on the natural gradient method. The physicochemical characteristics of the JcTIFY proteins were analyzed using the ExPASy website (<https://web.expasy.org/protparam/>). The subcellular localization of the proteins was predicted using the Plant mPLOC platform (<http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/>).

2.4. Analysis of gene structure, conserved domains, and protein motifs within the TIFY family of *J. curcas*

The gene structure of the JcTIFY family was examined using the Gene Structure Display Server 2.0 (<http://gsds.gao-lab.org/>). Additionally, TBtools (V2.001) was utilized to analyze the conserved domains of the JcTIFY protein family. MEME (V5.3.3, available at <https://meme-suite.org/meme/tools/meme>) was used to analyze conserved protein motifs within this family.

2.5. Cis-acting elements analysis of the TIFY family promoter in *J. curcas*

The JcTIFs encoding the region starting at the ATG and extending 2000 bp upstream were extracted using the PlantCARE website (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) for online analytical promoter region cis-elements, and the results were visualized with TBtools (V2.001).

2.6. Prediction of TIFY family interaction proteins in *J. curcas*

The protein sequences of JcTIFY from *J. curcas* were submitted to the STRING database (V11.5, <https://www.string-db.org>) for the interactive protein prediction of family members. Proteins with combined scores > 0.7 were screened, and interactions were displayed in Cytoscape V3.9.1, with family members illustrated using circles and lines. Proteins with strong interactions were labeled by color depth.

2.7. Expression profile of TIFY family genes in *J. curcas*

Transcriptome sequencing data were downloaded from the *J. curcas* database (JCDB) (Zhang et al., 2019) and normalized using log₂ (FPKM+1) (FPKM, fragments per kilobase million). Heat maps were generated using TBtools (V2.001), and the original data utilized are listed in Table S1.

2.8. RNA isolation and RT-qPCR analysis of the JAZ subfamily in *J. curcas*

Seedlings were exposed to 5°C and 5°C + 60 μmol·L⁻¹ MeJA, and the second and third true leaves of the seedlings were harvested for RT-qPCR analysis on days 0, 1, 2, 3, 4, and 5 of the treatment, respectively. Total RNA from *J. curcas* leaves was extracted according to the

instructions provided with the test kit (DP441) (Tiangen, Beijing, China) and reverse transcribed to synthesize the first strand of cDNA. Quantitative primers for JcJAZs were designed using the NCBI website, with GAPDH selected as the normalization reference gene. The primers utilized are detailed in Table S3. The experimental method for RT-qPCR was adapted from Kong et al. (2019), and data were calculated using the 2^{-ΔΔCT} method (Schmittgen and Livak, 2008). The original data used are listed in Table S2.

2.9. Interaction between members of the JAZ subfamily in *J. curcas*

The experimental method was conducted according to the approach described by Sheng et al. (2022), and the primers used are listed in Table S3. Negative and positive controls were included in each Petri dish, and images were captured after 3 days of incubation.

2.10. Luciferase complementation assay and subcellular localization

Luciferase complementation assay of JcTIFY10 was conducted as described by Sheng et al. (2022), with the primers detailed in Table S3. The subcellular localization of JcTIFY5A, JcTIFY9, and JcTIFY10 was carried out according to the method of Liang et al. (2023), with the primers used also listed in Table S3.

2.11. Statistical analysis

Each point in the figure represents the mean ± standard error (n = 3) and was statistically analyzed using one-way ANOVA followed by Duncan's test in SPSS V27. Different lowercase letters represent significant differences (*P* < 0.05).

3. Results

3.1. Identification and phylogenetic analysis of TIFY family members in *J. curcas*

Based on the Hidden Markov Model (PF06200), 33 proteins were screened from the protein repertoire of *J. curcas*. Using *A. thaliana* TIFY proteins as query sequences, the 65 best matches were selected using BLASTP. These candidate proteins were further subjected to a CD search in the NCBI database to verify specific conserved domains. By assembling their corresponding multiple transcripts encoded by the same gene, a total of 14 TIFY family members in *J. curcas* were identified (the CDS sequences and protein sequences used are listed in Table S4). The TIFY family in *J. curcas* is categorized into four subfamilies: TIFY, ZML, PPD, and JAZ based on phylogenetic analysis (Fig. 2). The TIFY subfamily consists of only one member (*JcTIFY8*). The ZML subfamily is comprised of four members: *JcTIFY1A*, *JcTIFY1B*, *JcTIFY2A*, and *JcTIFY2B*. The PPD subfamily contains two members: *JcTIFY4A* and *JcTIFY4B*. The JAZ subfamily is the largest, with seven members: *JcTIFY3*, *JcTIFY5A*, *JcTIFY5B*, *JcTIFY6*, *JcTIFY7*, *JcTIFY9*, and *JcTIFY10*. Additionally, *JcTIFYs* are grouped with *MeTIFYs*, which also belong to the

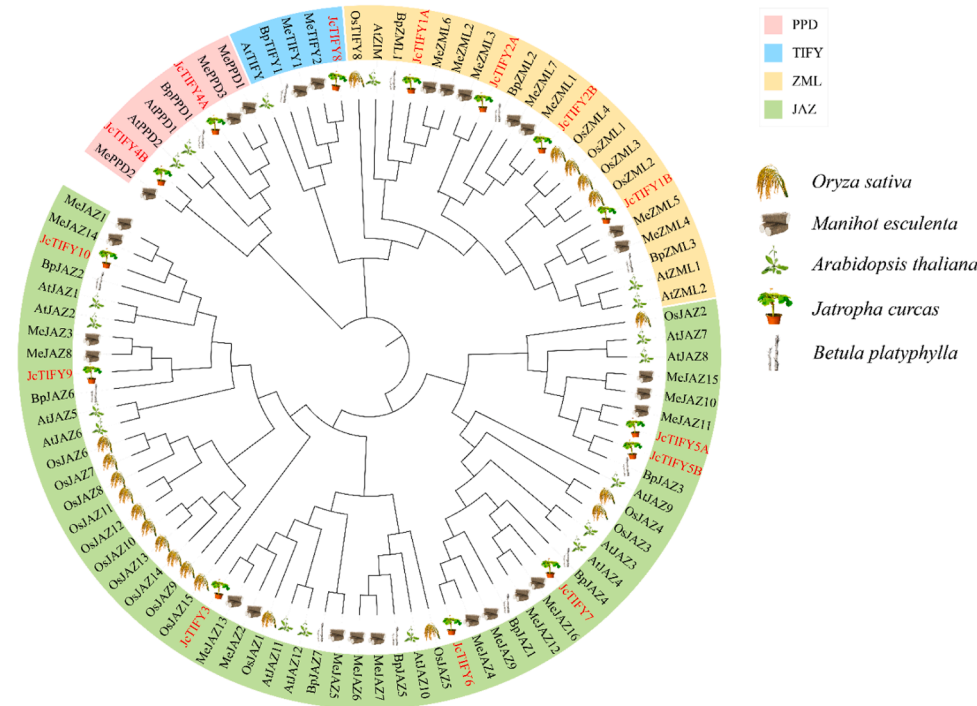


Fig. 2. Phylogenetic tree of TIFY members in *A. thaliana*, *O. sativa*, *M. esculenta*, *B. platyphylla* (black font) and *J. curcas* (red font). Different groupings are identified by various colors.

Euphorbiaceae family.

3.2. Analysis of physicochemical characteristics and chromosome localization of the TIFY family in *J. curcas*

The size of the TIFY proteins in *J. curcas* ranged from 15–48 kDa, with lengths of 135–458 amino acids. The instability coefficients of this protein family vary significantly. Members of the ZML subfamily were acidic, with isoelectric points (pI) ranging from 4.75–6.45, whereas the remaining family members were basic, with pI values ranging from 7.59–9.59. Consistent with studies in other species indicating that the TIFY family members act as transcription factors or repressors in the nucleus, *JcTIFYs* are also presumed to be located in the nucleus according to subcellular localization predictions. The hydrophilicity coefficients of all *JcTIFY* proteins were negative, indicating that they are all hydrophilic (Table 1). The chromosomal gene localization survey demonstrated that *JcTIFY9* was located on LG2 (linkage group, LG), *JcTIFY3/6/8/10* was located on LG11, *JcTIFY4B/4 A* was located on

LG8/9, *JcTIFY5A/5B/7* was located on LG3, *JcTIFY1A/2B* was located on LG4, and *JcTIFY1B/2 A* was located on LG10 (Fig. 3). Analysis of gene duplication events indicated that *JcTIFY1A/JcTIFY2B* and *JcTIFY5A/JcTIFY5B* arose from tandem duplications, whereas *JcTIFY3/JcTIFY9*, *JcTIFY10/JcTIFY9*, and *JcTIFY4A/JcTIFY4B* may have originated from segmental or whole-genome duplications. Ka/Ks values were calculated for the paralogous genes of *JcTIFY* in *J. curcas*, and the results showed that all values were less than 1, thus indicating purifying selection during evolution (Table S5).

3.3. Analysis of gene structure, conserved domains, and conserved protein motifs of the TIFY family in *J. curcas*

The TIFY family of *J. curcas* contains the conserved domains of TIFY, Jas, GATA, and CCT (Fig. 4). All these family members contain conserved TIFY domains. The CCT and GATA domains are unique to the ZML subfamily, and the GATA domain can specifically bind to the [AT] GATA[AG] promoter element of the conserved DNA sequence. Both the

Table 1
Characteristics of the TIFY proteins in *J. curcas*.

Gene name	Gene ID	Protein					
		Subcellular localization	Molecular weight/Da	Number of amino acid	Isoelectric point	Instability index	Hydropathicity
<i>JcTIFY9</i>	105631367	nucleus	28304.74	261	9.18	56.01	−0.764
<i>JcTIFY10</i>	105637994	nucleus	30360.01	281	9.46	62.89	−0.584
<i>JcTIFY3</i>	105638047	nucleus	22945.14	216	7.59	50.90	−0.410
<i>JcTIFY6</i>	105627964	nucleus	38282.18	363	9.59	48.20	−0.288
<i>JcTIFY7</i>	105634106	nucleus	39919.17	376	8.75	49.58	−0.251
<i>JcTIFY5B</i>	105634004	nucleus	16842.83	143	8.84	95.08	−1.101
<i>JcTIFY5A</i>	105634005	nucleus	15429.47	135	9.14	74.20	−0.825
<i>JcTIFY4B</i>	105649938	nucleus	35775.41	318	8.9	59.11	−0.914
<i>JcTIFY4A</i>	105632039	nucleus	43474.90	394	7.89	45.03	−0.610
<i>JcTIFY8</i>	105639251	nucleus	47797.90	458	8.96	45.88	−0.499
<i>JcTIFY2B</i>	105630725	nucleus	40066.43	367	4.77	46.46	−0.627
<i>JcTIFY2A</i>	105641446	nucleus	38498.58	347	4.75	48.02	−0.729
<i>JcTIFY1B</i>	105641445	nucleus	32456.04	305	5.48	42.91	−0.635
<i>JcTIFY1A</i>	105630724	nucleus	31049.18	287	6.45	38.85	−0.689

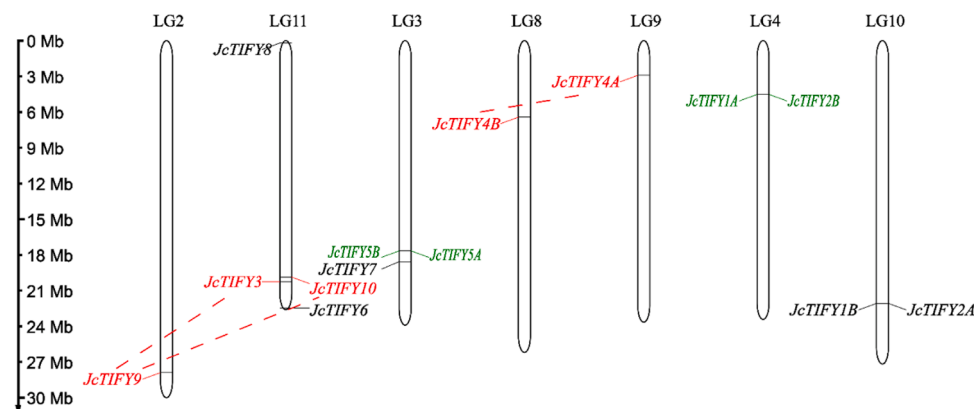


Fig. 3. Chromosome location and repeated events analysis of members of *JcTIFYs* in *J. curcas*. Note: LG2, LG3, LG4, LG8, LG9, LG10, and LG11 represent the seven linkage groups, and homologous pairs are presented in different colors and joined by lines. The scale bar on the left measures the length of the linkage groups (Mb).

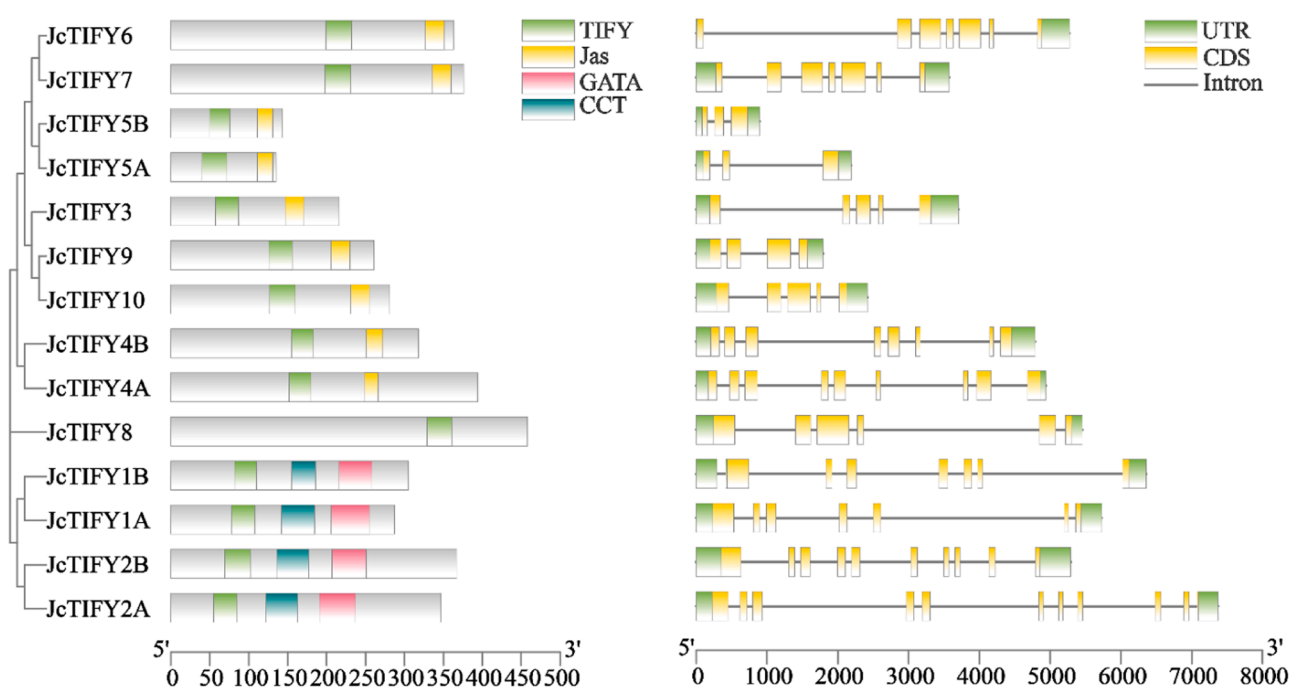


Fig. 4. Gene structure and conserved domains of *JcTIFY* proteins in *J. curcas*.

JAZ and PPD subfamilies have Jas domains; however, the PPD subfamily contains a Jas domain that lacks the PY motif. The TIFY family comprises only one conserved TIFY domain. The analysis of conserved motifs in TIFY family proteins was conducted using the MEME website, which identified seven conserved motifs. Motif 1 corresponded to the TIFY domain in all members. Motif 2 corresponded to the CCT and Jas domains with similar motifs and was present in all family members (Fig. S1). Using TBtools (V2.001) to analyze the gene structure, we observed that all 14 *TIFY* genes contained untranslated regions in both the N- and C-terminal regions. The gene structures varied significantly among the family members, and the intron-exon distribution patterns differed (Fig. 4). Sequence comparison of the protein conserved domains demonstrated that the conserved domains were highly conserved, with the sequences being the same among the subfamily members (Fig. S2).

3.4. Analysis of cis-acting elements of TIFY family promoters in *J. curcas*

The 2000 bp DNA sequence upstream of the *JcTIFY* start codon was submitted to Plant CARE for online analysis. The results showed that the cis-acting elements of the promoter regions of the 14 *JcTIFY* genes were

abundant and varied, indicating that *JcTIFYs* can be induced and expressed under various conditions and play roles in multiple signaling pathways involved in a variety of biological processes (Fig. 5). Each *JcTIFY* promoter region contains numerous photoresponsive and circadian regulatory elements, suggesting that light affects the expression of these genes. Moreover, they contain a large number of hormone response elements, including salicylic acid, JA, abscisic acid, gibberellin, and auxins, indicating that *JcTIFYs* are involved in multiple hormone signaling pathways. Almost all *JcTIFY* genes contained JA response elements with the exception of *JcTIFY8* and *JcTIFY2A*, indicating that these genes can respond to JA/MeJA signaling and regulate downstream gene expression. Additionally, the promoter regions of *JcTIFYs* also contain defense and stress response elements, indicating that *JcTIFYs* regulate the growth of *J. curcas* and significantly contribute to its response to both biotic and abiotic stresses.

3.5. Prediction of interacting proteins of the TIFY family in *J. curcas*

According to predictions from the STRING database and visualization of interaction results using Cytoscape software, *JcTIFY6* possessed

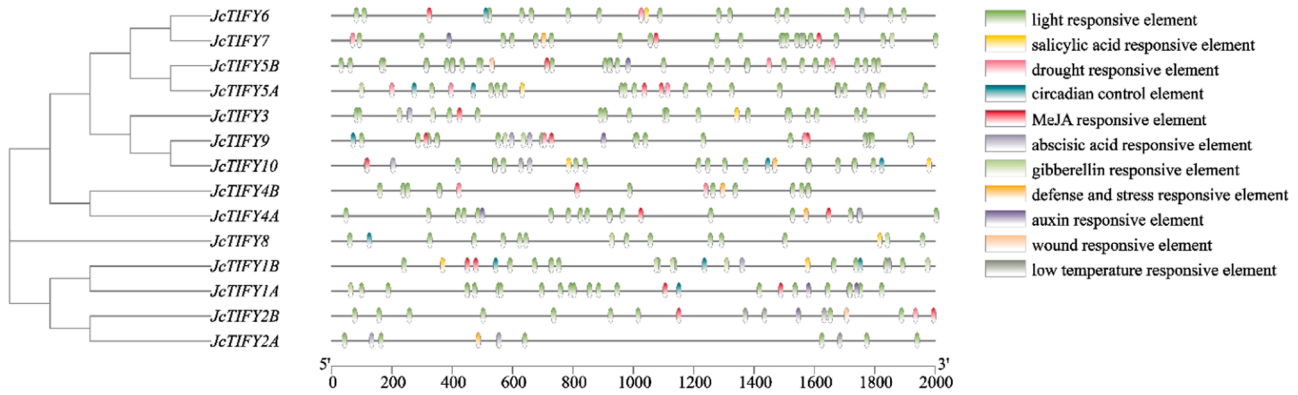


Fig. 5. Cis-acting elements analysis of *JcTIFYs* in *J. curcas*.

the most interacting proteins, while no interacting proteins were predicted for *JcTIFY2A/2B*. Most interacting proteins are family members, suggesting that heterogeneous complexes may form between TIFY proteins. The F-box protein *JcCOI1* interacted only with JAZ and exhibited a high combined score, whereas another F-box protein, SAP, was predicted to interact with *JcTIFY4B*. *A0A067K056*, *A0A067KDM9*, *A0A067JI90*, and *A0A067JMY5* are MYC transcription factors that interact most closely with JAZ proteins. *MTB1*, a transcription factor, was also predicted to interact with JAZ proteins and *JcTIFY1B*. *JcTIFY6* and *JcTIFY10* were predicted to interact with the ethylene response transcription factor *JcRAP2-7*, indicating that these two proteins are involved in the ethylene signaling pathway (Fig. 6).

3.6. Expression profile of the TIFY family in *J. curcas*

Using information from the JCDB database, changes in the expression of *JcTIFYs* under drought and salt stress, as well as in various organs, were analyzed to gain deeper insights into the function of TIFY family members in *J. curcas*. Under drought and salt stress, members of the same subfamily exhibited significant differences in expression. For example, under both drought and salt stress conditions, the expression of *JcTIFY2A* was lower, while the expression of other ZMLs (*JcTIFY1A*, *JcTIFY1B* and *JcTIFY2B*) was higher (Fig. 7A–D). Similarly, the expression patterns of the same gene in the roots and leaves of *J. curcas* under

salt stress were notably different. For example, *JcTIFY6* showed an expression pattern that first increased and then decreased in the roots, whereas it exhibited a continuous increase in the leaves (Fig. 7C,D). Furthermore, there were significant differences in the expression of *JcTIFYs* within the same organs and between the same *JcTIFY* in various organs. Members of the JAZ subfamily (with the exception of *JcTIFY5A/5B*) showed high expression in all organs, indicating that the JA signaling pathway plays key roles in all organs and affects plant growth and morphogenesis (Fig. 7E). In the PPD subfamily, the expression of *JcTIFY4B* was high in all organs, whereas that of *JcTIFY4A* was relatively low. Additionally, *JcTIFY8* was expressed in all organs. The expression of *JcTIFY2A* in the ZML subfamily was low, and it was only expressed in seeds, whereas the expression levels of other ZML members were high in the plants. These data suggest that the JcTIFY proteins regulate plant growth, organ development, and stress resistance (Fig. 7E).

3.7. Effects of MeJA on the expression of JAZ subfamily genes in *J. curcas* under chilling stress

As presented in Fig. 8, the expression of *JcJAZs* in *J. curcas* increased several-fold during a chilling treatment lasting 5 days, and the expression levels were further increased after the exogenous application of MeJA. Among them, *JcTIFY9*, *JcTIFY10*, and *JcTIFY5B* exhibited the most pronounced reactions to low temperature and exogenous MeJA. The expression of JAZ members (with the exception of *JcTIFY6*) showed an increasing and then decreasing trend during the 5-day chilling treatment, ultimately reaching a peak value on the 3rd day (*JcTIFY9*) or 4th day (*JcTIFY3*, *JcTIFY5A*, *JcTIFY5B*, *JcTIFY7*, and *JcTIFY10*). The exogenous addition of MeJA increased the expression of JAZs and advanced their expression peaks. Interestingly, *JcTIFY5B* expression peaked on the first day of treatment and then decreased, indicating that exogenous MeJA enhanced the expression of *JcJAZs* under chilling stress.

3.8. Interactions between members of the JAZ subfamily in *J. curcas*

First, the auto-activation activity of the *JcJAZs* was verified. The hybrid offspring of yeast Y187 (pGADT7-*JcJAZs*) and yeast Y2Hgold (pGBKT7), as well as the hybrid offspring of yeast Y187 (pGADT7) and yeast Y2Hgold (pGBKT7-*JcJAZs*), were plated onto SD-Trp-Leu (DDO) medium and SD-Trp-Leu-His-Ade/X- α -Gal (QDO/X) medium. It was observed that the diploid hybrids could grow on DDO medium but not on QDO/X medium, indicating that *JcJAZs* did not possess auto-activation activity (Fig. 9A). The hybrid offspring of yeast Y187 (pGADT7-*JcJAZs*) and yeast Y2Hgold (pGBKT7-*JcJAZs*) were then plated onto DDO medium and QDO/X medium to verify the interactions among *JcJAZs*. According to the results of the yeast dihybrid assays, homodimers or heterodimers can form among the JAZ subfamily members. Among these, *JcTIFY3*, *JcTIFY6*, and *JcTIFY10* can form

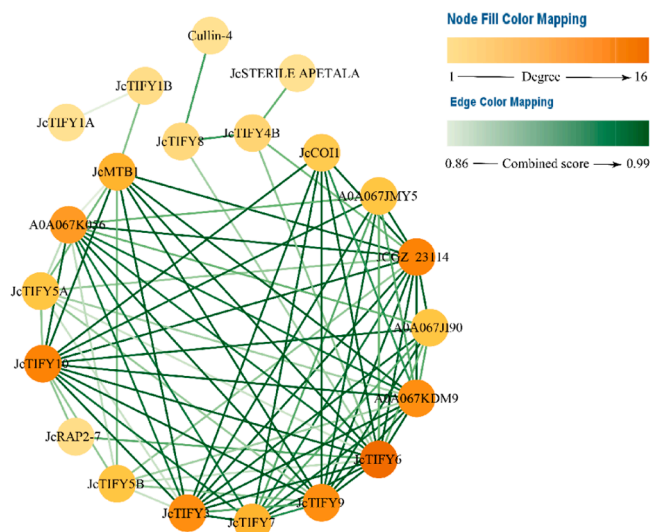


Fig. 6. Network analysis of *JcTIFYs* and interacting proteins in *J. curcas*. Note: The depth of the circle color indicates the number of interacting proteins predicted by *JcTIFY* proteins. The depth of the line color indicates the strength of the interactions between the proteins.

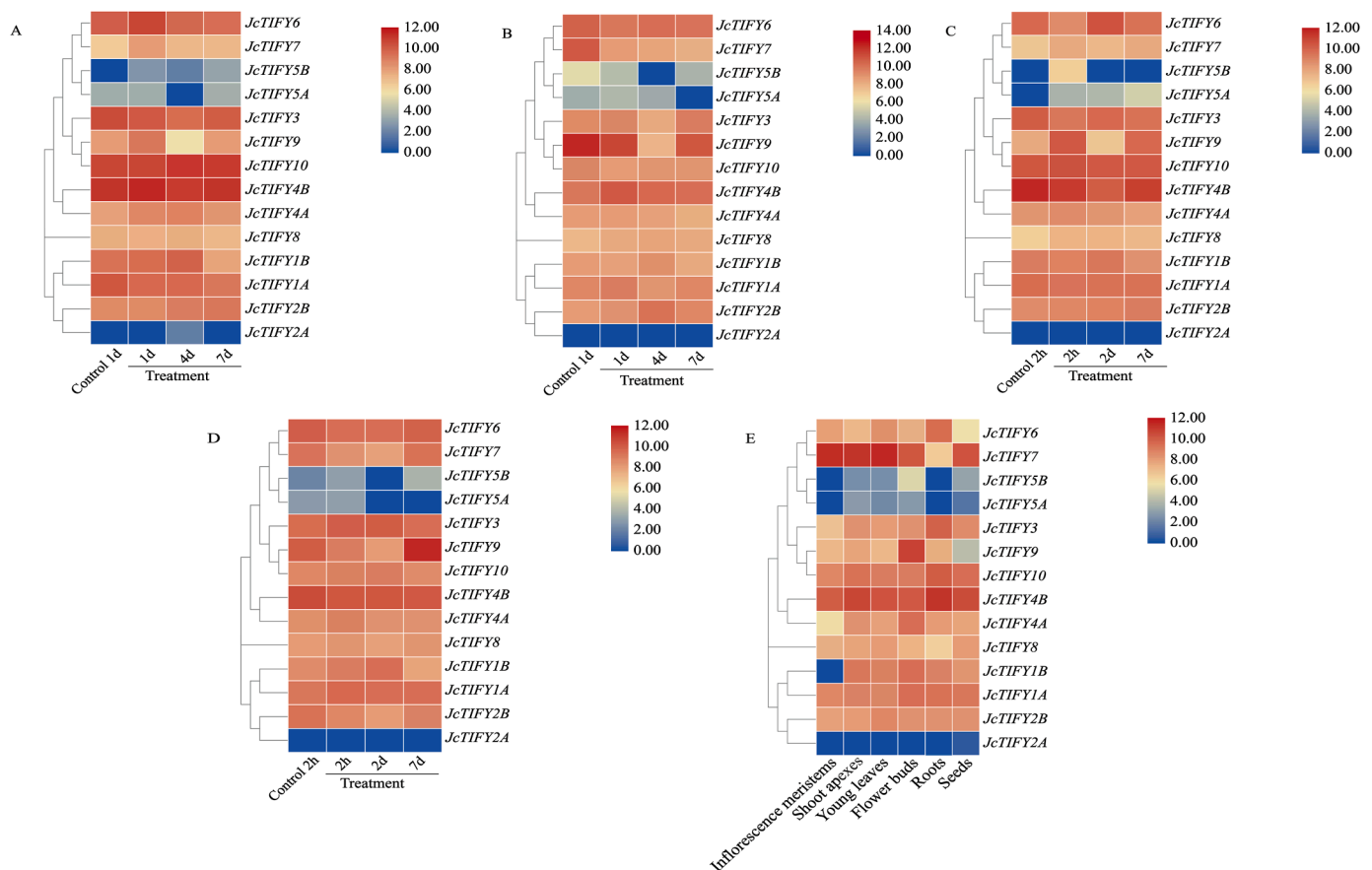


Fig. 7. Analysis of the *JcTIFY* gene expression profile in *J. curcas* under drought stress, salt stress and in various organs (A: roots, drought stress; B: leaves, drought stress; C: roots, salt stress; D: leaves, salt stress; E: different organs).

homodimers, whereas *JcTIFY3/10*, *JcTIFY5A/7*, *JcTIFY5A/9*, *JcTIFY5A/10*, *JcTIFY5B/6*, *JcTIFY5B/7*, *JcTIFY5B/10*, *JcTIFY6/7*, *JcTIFY6/10*, *JcTIFY7/10*, and *JcTIFY9/10* can form heterodimers. This suggests that *JcJAZ* proteins likely function by forming dimers (Fig. 9B).

3.9. *JcTIFY10* can form homodimers *in vivo*

To further examine the interactions between *JcTIFYs*, we constructed the *JcTIFY10*-pCambia1300-Nluc and *JcTIFY10*-pCambia1300-Cluc vectors for the LCA, respectively. No fluorescence was observed in the control group, whereas the combination of *JcTIFY10*-pCambia1300-Nluc and *JcTIFY10*-pCambia1300-Cluc emitted fluorescence, further indicating that *JcTIFY10* forms homodimers *in vivo* (Fig. 10).

3.10. TIFY domain is indispensable for the formation of homodimers and heterodimers of *JcTIFYs*

To test whether the Jas domain and the TIFY domain of the *JcJAZ* proteins are responsible for the interaction with each other, we constructed truncated versions of *JcTIFY10* lacking the Jas and TIFY motifs (*JcTIFY10ΔJas* and *JcTIFY10ΔTIFY*, respectively). They were then combined with pGADT7 for the yeast two-hybrid assay. The results are presented in Fig. 11, where *JcTIFY10* can interact with *JcTIFY3*, *JcTIFY5A*, *JcTIFY5B*, *JcTIFY6*, *JcTIFY7*, *JcTIFY9* and *JcTIFY10* to form dimers. *JcTIFY10ΔJas* yielded the same results, but *JcTIFY10ΔTIFY* could not interact with these proteins. This demonstrates that the TIFY motif mediates homomeric and heteromeric interactions between *JcJAZs*.

3.11. Subcellular localization of *JcTIFY5A*, *JcTIFY9*, and *JcTIFY10*

Determining the subcellular localization of *JcTIFY* is essential for elucidating its function. *JcTIFYs* were predicted to be localized in the nucleus. The results demonstrated that *JcTIFY5A*, *JcTIFY9*, and *JcTIFY10* proteins were indeed localized in the nucleus. Meanwhile, *JcTIFY9* was also observed to exhibit cytoplasmic localization in addition to its nuclear localization (Fig. 12).

4. Discussion

TIFY proteins, which are only present in terrestrial plants, play crucial roles in plant development and growth, as well as in their responses to stress (Vanholme et al., 2007). There have been several reports regarding the TIFY family in other plants, including tomato and *A. thaliana*, with in-depth studies on the functions of some of its members (Swinnen et al., 2022; Hakata et al., 2017). *J. curcas* is an oil-bearing plant of tropical origin, and various stresses negatively affect its growth and industrial expansion. Studying *JcTIFYs* is conducive to understanding their role in abiotic stress responses and providing new insights for breeding *J. curcas* with enhanced resistance. In the present study, the entire genome of *J. curcas* was analyzed, and 14 members of the TIFY family were identified. These 14 members are located on seven linkage groups, with LG11 containing four family members (*JcTIFY3/6/8/10*). However, LG2, LG8, and LG9 each contain only one member, including *JcTIFY9*, *JcTIFY4B*, and *JcTIFY4A*, respectively. Furthermore, collinearity analysis identified the presence of two tandem duplicate gene pairs and three segmental or whole-genome duplication gene pairs among the *JcTIFYs*.

The TIFY family was divided into four subfamilies (ZML, JAZ, PPD,

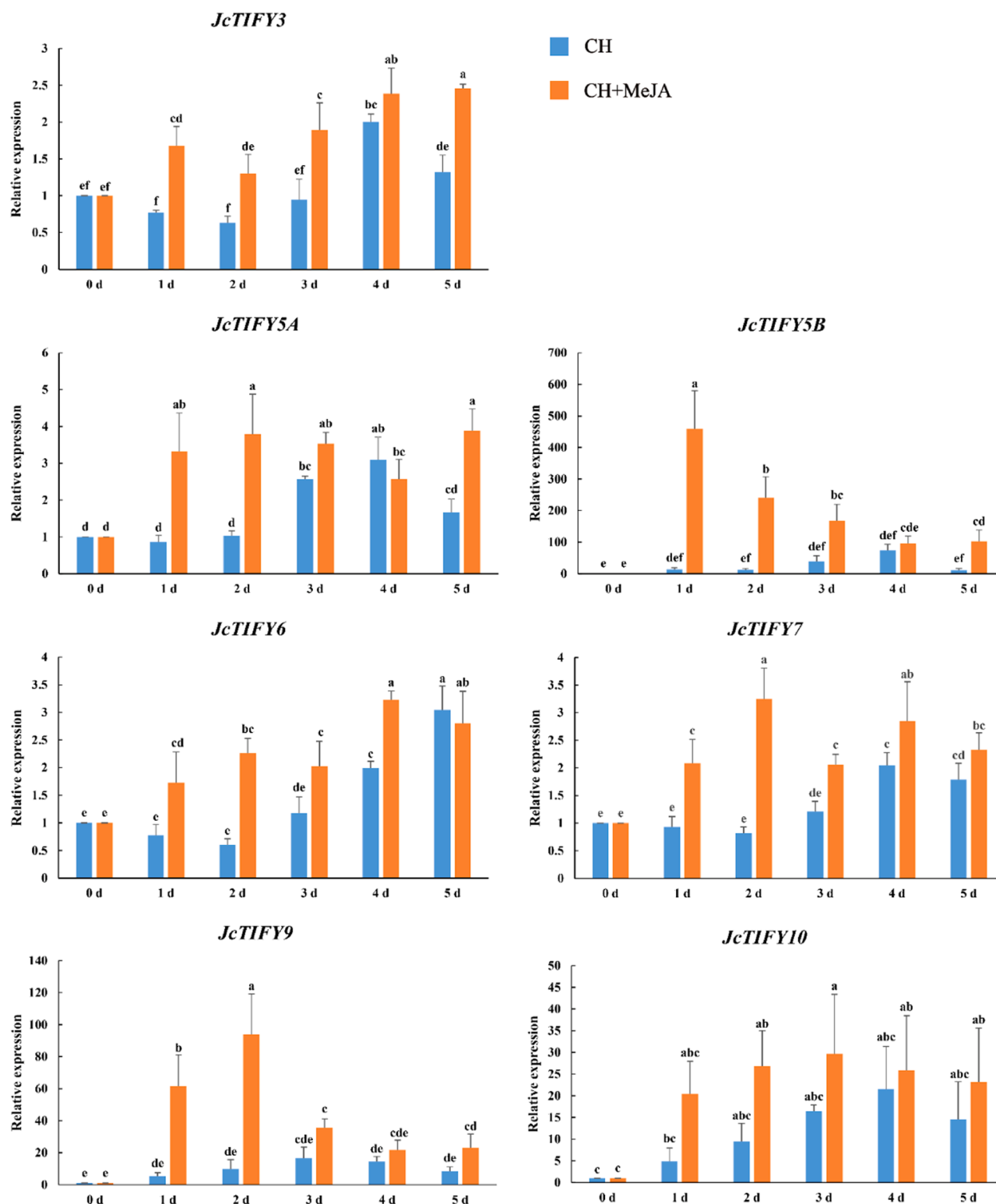


Fig. 8. RT-qPCR analysis of JcTIFY expression in *J. curcas* under different treatments (CH; 5°C; CH+MeJA; 5°C+60 $\mu\text{mol}\cdot\text{L}^{-1}$ MeJA). Note: Different lowercase letters represent significant differences among treatments ($P < 0.05$).

and TIFY) based on phylogenetic research (Tao et al., 2022). The JcTIFYs included seven JAZ, two PPD, one TIFY, and four ZML members, respectively. JcTIFY members of the same subfamily possessed the same conserved domains. All members contain a highly conserved TIFY domain located at the center of the protein. JcTIFY proteins vary significantly in molecular weight, ranging from 15–48 kDa, and all JcTIFYs are hydrophilic proteins located in the nucleus. The subcellular

localization of JcTIFY5A, JcTIFY9, and JcTIFY10 was consistent with the predicted results. JcTIFY9 was also observed to exhibit cytoplasmic localization in addition to nuclear localization. Genetic structure analysis demonstrated that JcTIFYs are highly conserved evolutionarily, and the genetic distribution and number of introns differed, even among members from the same evolutionary branch. All family members possessed motif 1 (TIFY), and motif 1 of the same evolutionary branch

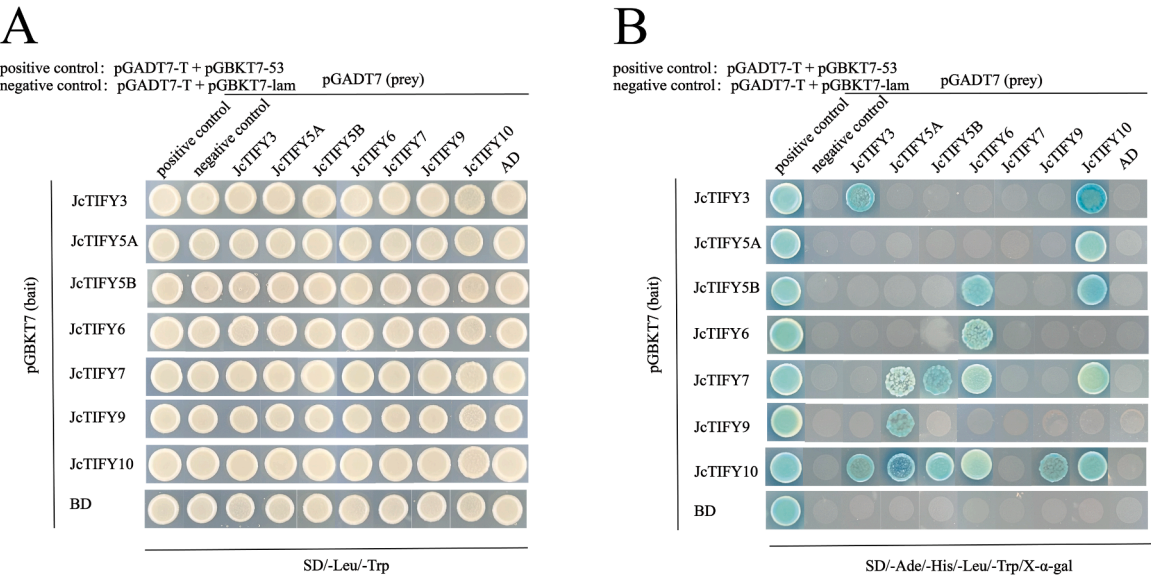


Fig. 9. Protein-protein interactions between JcJAZ proteins. A: Hybrid yeast cells that tested positive were cultivated in DDO medium. B: X-α-Gal was added to the QDO medium to evaluate the interactions.

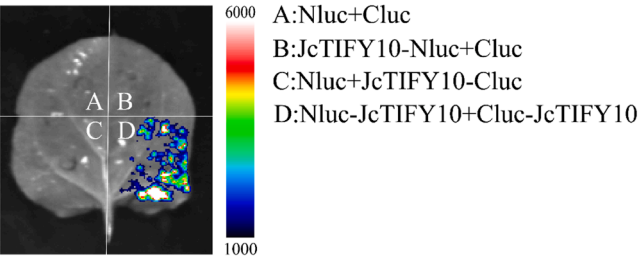


Fig. 10. Split firefly luciferase complementation assay confirmed that JcTIFY10 formed homologous dimers.

was distributed in similar locations. These results indicate significant differences in gene structure, protein coding, and evolutionary relationships among the different groups of the JcTIFY family. The first 2000 bp sequence of the gene family members contained numerous cis-acting elements associated with plant growth, development, and

stress resistance, thus indicating that the gene family is crucial for abiotic stress adaptation in *J. curcas*.

As an important transcriptional suppressor, the JAZ subfamily regulates plant physiological processes by binding to or releasing transcription factors, such as MYC (Major et al., 2017). The exogenous addition of MeJA induces the accumulation of JA-Ile (an active form of jasmonate), prompting COI1 to bind the JAZ protein. The JAZ protein is then degraded via ubiquitination by the SCF^{COI1} complex, ultimately releasing the MYC2 transcription factor, which in turn promotes JAZ transcription by binding to the JAZ promoter region (Chini et al., 2007). When the expression levels of JAZs are sufficiently increased, they inhibit the transcriptional activation function of downstream transcription factors, thereby restoring the JA signaling pathway to an inactive state. This indicates that MYC2 regulates the expression of JAZs (Chini et al., 2007; Lu et al., 2020). Low temperature can also significantly upregulate the expression of some JAZs (Zhang et al., 2015), inferred from the observation that cold treatments promote the production of endogenous JA in plants (Hu et al., 2013). In the current research, JAZ expression first increased and then decreased over time

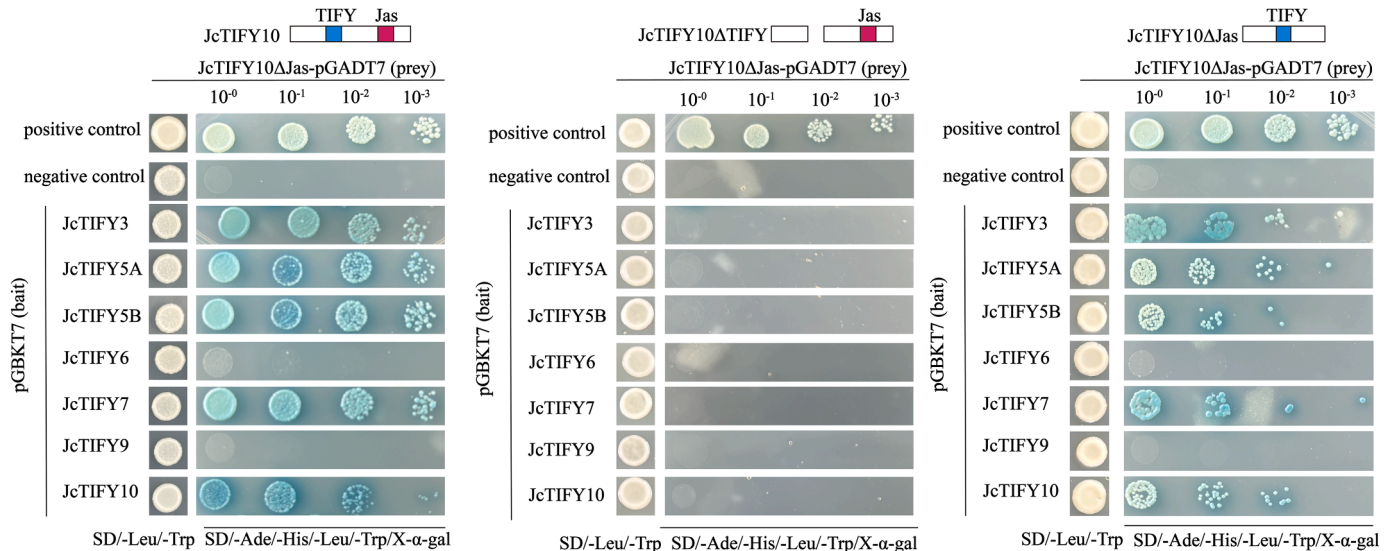


Fig. 11. Interaction of JcTIFY10 and its variants with JcJAZs.

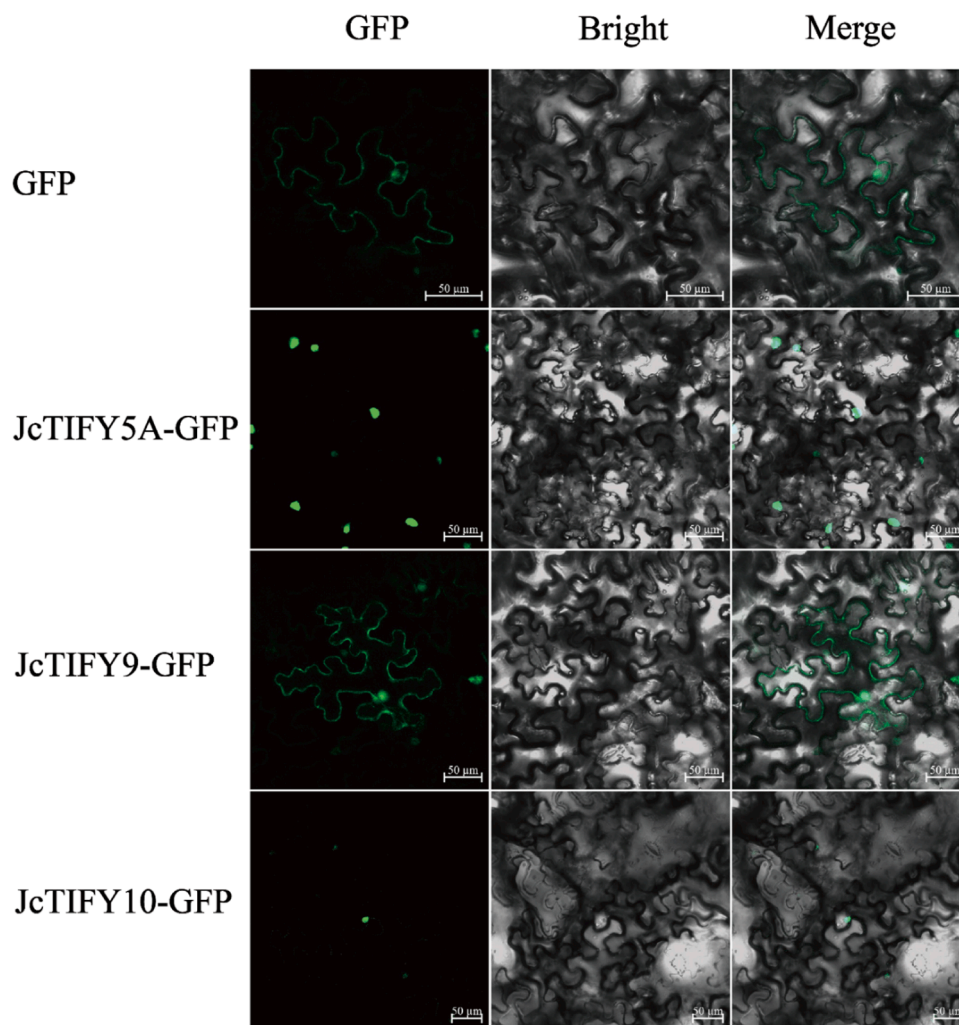


Fig. 12. Analysis of subcellular localization of JcTIFY5A, JcTIFY9, and JcTIFY10.

under low-temperature stress, whereas exogenous MeJA advanced and improved this process, indicating that *JcJAZs* respond to cold stress and exogenous MeJA treatment. Additionally, salt and drought stresses can induce the expression of *JcTIFYs*, suggesting that the cis-acting elements of *JcTIFYs* related to stress resistance may play a crucial role in this process.

JAZs can form homodimers or heterodimers through the TIFY domain in *A. thaliana* (Chini et al., 2009), and our study also confirmed that the TIFY domain is indispensable for the formation of homodimers and heterodimers of JcTIFYs. It has been reported in peaches (Sheng et al., 2022) and cotton (Li et al., 2017) that JAZ proteins can interact to form homologous or heterodimers, and the dimerization enhances the stability of JAZ proteins (He et al., 2020). In Arabidopsis, some JAZ proteins form homodimers or heterodimers with other JAZ proteins, including AtJAZ2, AtJAZ5, AtJAZ10, etc. (Chini et al., 2009). In this study, using software prediction and yeast two-hybrid experiments, we observed that homodimers or heterodimers could also be formed between JcJAZs; however, the predicted results were partially different from the experimental results. Unlike *A. thaliana*, all members interact with other JAZ proteins to form heterodimers in *J. curcas*. Among these, JcTIFY10 interacted with all other JAZ proteins *in vitro* and formed a homodimer *in vivo*, indicating that the stability of JcTIFY10 is crucial for JA/MeJA signaling.

5. Conclusion

In this study, TIFY family genes in *J. curcas* were identified and analyzed, revealing 14 JcTIFY family members classified into four subfamilies (ZML, JAZ, PPD, and TIFY). Various characteristics, interactions, and evolutionary connections of the JcTIFY proteins, as well as gene structures and expression profiles, were discussed. Under low-temperature stress, the expression of *JcJAZs* showed an initial increase followed by a decrease, and the expression of *JcJAZs* was regulated by MeJA. Notably, MeJA treatment upregulated the expression of *JcTIFYs* under low-temperature stress, with JcTIFY5B being the most significantly upregulated. Interactions among JcJAZs were evident in yeast two-hybrid assays, while JcTIFY10 was shown to form homodimers both *in vivo* and *in vitro*. Our experiments confirmed that JcJAZ proteins can form heterodimers with other JcJAZ proteins, and some proteins can also form homodimers through the TIFY structural domain. Subcellular localization showed that JcTIFY5A and JcTIFY10 were located in the nucleus, while JcTIFY9 was located in both the cytoplasm and nucleus. These findings enhance our understanding of TIFY gene-mediated responses to low-temperature stress in *J. curcas*.

CCRediT authorship contribution statement

Yunliang Chen: Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Xingyuan Liu:** Validation. **Yanfei Bai:** Validation. **Shanshan Lan:** Software, Methodology,

Investigation, Conceptualization. **Shuang long Yang**: Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization. **Jin Sha**: Validation. **Zhurong Zou**: Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.indcrop.2024.120194](https://doi.org/10.1016/j.indcrop.2024.120194).

Data Availability

Data will be made available on request.

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